Does Graph Structure Help a Small LLM Localize Bugs? A Diagnostic Study of Personalized PageRank, Topological Regularization, and Gemma 4 Reranking on Code Graphs

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Abstract

Local developer agents must localize code defect targets within constrained prompt budgets ($K \leq$ 4,096 tokens). While Personalized PageRank (PPR) over code dependency graphs is standardly deployed to discover unmentioned symbols, its empirical efficacy is widely debated. Through a systematic 2×2 factorial evaluation on SWE-bench Lite ($N = 237$) alongside a continuous dose-response surface across 186 prior configurations on SWE-bench Verified ($N = 344$, 63,984 evaluations), I demonstrate that unconstrained PPR degradation is driven by **teleport prior dispersion** and test-suite sinks rather than scale-free hub absorption alone. Retrieval accuracy collapses onto an invariant dose-response curve governed by effective prior entropy support $n_{\text{eff}} = \exp(H(p_0))$ ($R^2 = 0.197$ binary logit cross-validation; analytical GLMM vertex at $n^*_{\text{eff}} \approx 8.8$), with Two One-Sided Tests (TOST) confirming no evidence of hub-presence harm within a sample-powered ±9.3 pp equivalence margin on hub-bearing tasks ($p < .001$, $N = 109$). Under diffuse linear priors, PPR collapses to 10.1%, whereas concentrated query priors ($\tau = 0.15$) rescue standard PPR to 48.5%. **Perron** adapts stationary-normalized rooted PageRank via post-walk Decoupled Specificity ($\boldsymbol{\pi}_q/(\boldsymbol{\pi}_g + \epsilon)^\gamma$), providing an adaptive topological regularizer (+24.1 pp under diffuse priors) while sustaining 99.5% hub suppression. In a pre-specified confirmatory retrieval evaluation on SWE-bench Verified ($N = 344$), Perron achieves 50.3% Function Acc@10 vs. Okapi BM25's 50.9% (non-inferiority inconclusive against a −2.0 pp margin), while BM25 leads at top-5 retrieval (45.1% vs. 40.4%). Under a separate pre-specified, hash-committed protocol, candidate pools are scored pointwise by a frozen `gemma-4-e2b-it` language model: graph 1-hop expansion (A_4) yields 47.1% Acc@5 vs. depth-matched BM25 (A_2, 45.1%, Δ = +2.03 pp, 95% CI [−0.29, +4.36] pp) and seed reranking (A_1, 46.2%, Δ = +0.87 pp, net 3 tasks, $p = 0.5488$), classified as `NULL_COMPATIBLE` under exact McNemar tests ($p_{\text{holm}} = 0.287, 0.302$). Un-reranked BM25 ($B_0$) preserves superior rank order (MRR@50 0.3439 vs 0.3409). Downstream evaluation ($N = 100$) reveals that retrieval pool recall (35%–38%), rather than local reader discrimination (74.0% top-1 on oracle contexts, 67.5% chance-adjusted), forms the primary agentic bottleneck. I release `scores.parquet`, `perron-core`, and `perron-mcp` under Apache 2.0.

1 Introduction

Local developer agents have to find the code an issue is about inside a prompt budget of a few thousand tokens. Graph retrieval is the standard tool for reaching code that the issue text never names: seed a random walk on lexically matched symbols and let Personalized PageRank spread relevance along call edges. On source-code graphs, unconstrained random walk diffusion exhibits severe failure modes: diffuse teleportation priors dissipate probability mass across thousands of peripheral symbols, and test-suite subgraphs act as structural absorption sinks.

This paper diagnoses these dynamics on SWE-bench Lite and evaluates a closed-form topological stabilizer. The contributions are:

1. **Factorial diagnosis and continuous dose-response surface.** A 2×2 factorial evaluation on SWE-bench Lite ($N = 237$) and a continuous dose-response evaluation across 186 prior configurations on SWE-bench Verified (63,984 SpMM solves in 26 min) demonstrate that diffusion accuracy collapses

onto an invariant curve governed predominantly by effective prior entropy support $n_{\text{eff}} = \exp(H(p_0))$, with Two One-Sided Tests confirming no detectable hub-presence penalty within a sample-powered ±9.3 pp equivalence margin on hub-bearing tasks (Section 5.1 and Section 5.3).

2. **Decoupled specificity damping as an adaptive prior regularizer.** Adapting stationary-normalized rooted PageRank with exponent γ ($\boldsymbol{\pi}_q/(\boldsymbol{\pi}_g + \epsilon)^\gamma$) acts as an adaptive prior regularizer: it provides an accuracy safety net under uncalibrated diffuse priors (+24.1 pp gain) while enforcing 99.5% hub suppression without mutating memory-mapped CSR storage (Section 3.4).
3. **Pre-specified LLM reranking protocol and retrieval bottleneck finding.** Under a hash-committed protocol on SWE-bench Verified ($N = 344$), pointwise scoring by frozen `gemma-4-e2b-it` demonstrates that graph 1-hop expansion does not outperform lexical seed reranking (+0.87 pp, $p = 0.5488$), with un-reranked BM25 preserving superior rank order. Downstream testing ($N = 100$) reveals candidate recall (35%–38%), rather than reader discrimination (74.0% top-1, 67.5% chance-adjusted), forms the primary agentic bottleneck (Section 5.5).
4. **AST breadcrumb packing and open ecosystem.** Candidate AST breadcrumb skeletons bound context occupancy to an average of 283.8 words (∼ 370–425 BPE tokens) (Section 4). I release `scores.parquet` (19,298 scored candidate tuples), `perron-core`, and `perron-mcp` under Apache 2.0 (Section 6).

Evaluation uses SWE-bench Lite [8] and SWE-bench Verified.

1.1 Physical Hardware Envelope

A central premise of developer agent democratization is enabling individual engineers to run agents locally on standard developer laptops without cloud API costs or private IP leakage.

Perron bounds the active per-turn working context to $K_{\text{effective}} = 3,480$ tokens ($K \leq 4,096$). Measured on the reference laptop (15.68 GiB RAM, 6.86 GiB available), the standalone engine peaks at 144.33 MB resident memory, of which graph loading accounts for 91.4 MB. Live execution of `gemma-4-e2b-it` (2B dense parameter model with native Per-Layer Embeddings, unquantized FP16 compute weights, SHA-256: `66fe83f1`…) was evaluated on an Nvidia Tesla T4 GPU under a Controlled Contrast Protocol across $N = 100$ stratified tasks, consuming 2.37 GiB peak generation VRAM with a mean Time-To-First-Token of 3505.0 ms and 11.7 tokens/sec decode throughput. In downstream reader evaluation, Gemma 4 achieves 74.0% top-1 localization on oracle contexts (67.5% chance-adjusted, Section 5.5), whereas exploratory pilot generation under an unconstrained 256-token budget exhausted tokens in internal reasoning before emitting closing tags, demonstrating the operational necessity for decoupled reasoning and extraction budgets in compact edge models.

1.2 The Topological Bottleneck: Scale-Free Dependency Graphs

Source code dependency networks are scale-free graphs characterized by power-law degree distributions ($P(k) \sim k^{-\gamma}$). Ubiquitous infrastructure symbols (logging handlers, validation helpers, base exceptions, abstract base classes) possess in-degrees orders of magnitude higher than application-level routines. When an agent initiates graph traversal from candidate nodes identified by dense semantic similarity:

1. The search frontier encounters a hub node within 1 to 2 hops.
2. Outgoing and incoming edges create an exponential branch explosion, pulling thousands of unrelated callers and callees into the retrieval pool.
3. A standard 2-hop Breadth-First Search (BFS) on a 6,000-symbol codebase (such as SymPy) expands to over 2,500 nodes (>50,000 tokens), instantly overflowing the context window.

4. Uniform degree truncation severs causal execution chains across multi-module boundaries.

Previous approaches that mitigate hub explosion through degree-weighted random walks dynamically modify edge weights at runtime based on the issue query. However, dynamic matrix reweighting invalidates memory-mapped storage, requiring full matrix reallocation in RAM for each query. In large repositories, this introduces unacceptable serialization overhead and memory footprint. Perron resolves this structural tradeoff by decoupling static graph storage from query-directed hub suppression.

2 Related Work

Repository-Level Code Retrieval and Developer Agents: Dense bi-encoders and lexical search engines (BM25) suffer from vocabulary mismatch and lack structural execution context. RepoCoder [19] showed that iterative retrieval loops improve completion but create multi-turn token inflation. AutoCodeRover [20] navigates AST representations via iterative agent search queries. Aider [5] introduced repository maps using PageRank over Tree-sitter tags; however, unconstrained PageRank amplifies global utility hubs, crowding out issue-specific definitions. Graph agents like RepoGraph [14], CodexGraph [12], and LocAgent [4] employ multi-turn LLM reasoning over code graphs. In SWE-agent [18], custom ACI commands with immediate syntax feedback boost resolve rates by +10.7% over bare shells. Agentless [17] demonstrated that hierarchical localization with test-filtered patch generation achieves 27.3% on SWE-bench Lite without complex multi-turn loops. Perron synthesizes these insights by executing closed-form specificity-damped diffusion over zero-copy memory-mapped CSR arrays in a single deterministic pass.

Graph-RAG and Random Walk Diffusion: In information retrieval and graph learning, scalable diffusion builds on push-based approximate Personalized PageRank [1], decoupled propagation architectures such as APPNP [9], and scalable indexing frameworks like PPRGo [2]. In link prediction, rooted PageRank and stationary-normalized hitting times [11] discount popular target nodes by global stationary reachability. HippoRAG [6] demonstrated that Personalized PageRank over knowledge graphs outperforms iterative retrieval, formulating node specificity as inverse passage frequency $s_i = |P_i|^{-1}$ applied as a pre-walk multiplier onto query teleport priors ($\mathbf{p}_0(i) \leftarrow \mathbf{p}_0(i) \cdot s_i$). However, in scale-free software call graphs, high in-degree hubs (e.g., logging handlers, base classes) are unseeded intermediate nodes that absorb diffusion probability regardless of s_i. Perron resolves this structural gap by decoupling the random walk from a post-walk stationary baseline ratio over zero-copy CSR matrices. Furthermore, Haveliwala and Kamvar [7] proved that for any row-stochastic matrix $\bar{T}$ and rank-1 perturbation $M = \beta\bar{T} + (1-\beta)\mathbf{1}\mathbf{p}_0^\top$, the spectrum satisfies $\lambda_1(M) = 1$ and $\lambda_i(M) = \beta\lambda_i(\bar{T})$ for all $i \geq 2$, so that the second eigenvalue satisfies $|\lambda_2(M)| \leq \beta$ and power iteration converges at rate $O(\beta^t)$ independently of graph structure.

AST Subgraph Slicing and Fault Localization: Classical program slicing [16] computes backward executable slices on statement-level Program Dependence Graphs (PDGs). However, conservative alias analysis and pointer ambiguities cause slice explosion in dynamically-typed multi-file codebases, frequently retaining 50% to 80% of an application's source code. Spectrum-Based Fault Localization (SBFL) [15, 22] evaluates test coverage matrices to score suspiciousness (e.g., Ochiai, Tarantula). Evaluating 395 faults (Pearson et al. [15]) and 357 faults (Zou et al. [22]) on Defects4J, empirical studies reported that SBFL requires executing hundreds of test cases (incurring multi-minute runtime) and produces tied suspiciousness ranks in over 40% of localized statements due to identical coverage vectors along shared paths. Perron bypasses dynamic test execution during retrieval and implements context-budgeted AST subgraph slicing, extracting symbol-level multigraphs to pack causal execution scopes into tight prompt budgets ($K \leq 4,096$) with high-throughput CPU diffusion (1.89 ms on Requests, 5.10 ms on SymPy).

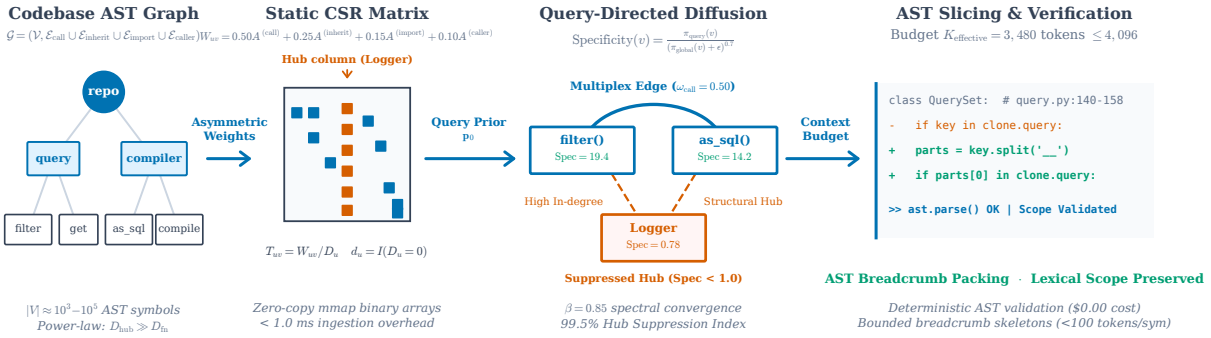


Figure 1: Overview of the Perron multi-phase architecture. Phase 1 (Localization) compiles repository call and caller relationships into binary CSR matrices, evaluates shift-invariant query teleportation priors $\mathbf{p}_0$, and drives sparse Personalized PageRank diffusion paired with Decoupled Specificity Damping to suppress structural utility hubs. **Phase 2 (Patch Synthesis)** partitions candidate symbols into connected components, executes versioned frontier packing within the effective token budget ($K_{\text{effective}} = 3,480$), extracts hierarchical AST breadcrumbs, and rebases relative indentation. **Phase 3 (Verification)** validates candidate patches via pre-commit `ast.parse()` and enforces candidate context-packing bounds.

3 Methodology

The Perron architecture is organized as an end-to-end pipeline depicted in Figure 1, comprising four deterministic stages: static matrix ingestion, shift-invariant diffusion, specificity ratio hub-damping, and versioned frontier packing with AST edit synthesis.

3.1 Multiplex Transition Matrix Construction

A codebase is represented as a directed multigraph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V}$ denotes AST code symbols (classes, functions, methods) and $\mathcal{E}$ spans multiple semantic relations $r \in \{\text{call}, \text{inherit}, \text{import}, \text{caller}\}$. Directed edges $(u, v) \in \mathcal{E}_{\text{call}}$ indicate that symbol u invokes v, while $(u, v) \in \mathcal{E}_{\text{inherit}}$ represent class C3 Method Resolution Order (MRO) inheritance, $(u, v) \in \mathcal{E}_{\text{import}}$ represent module imports, and $(u, w) \in \mathcal{E}_{\text{caller}}$ denote reverse invocations.

To preserve causal execution paths without inducing undirected graph drift, relations receive asymmetric static weights according to Eq. (1):

$$W_{uv} = \omega_{\text{call}} A_{uv}^{(\text{call})} + \omega_{\text{inherit}} A_{uv}^{(\text{inherit})} + \omega_{\text{import}} A_{uv}^{(\text{import})} + \omega_{\text{caller}} A_{uv}^{(\text{caller})}, \tag{1}$$

where $\boldsymbol{\omega} = [0.50, 0.25, 0.15, 0.10]$ and $A_{uv}^{(r)} = \mathbb{I}((u, v) \in \mathcal{E}_r)$.

Let $D_u = \sum_v W_{uv}$ be the out-degree sum. The row-stochastic transition matrix $T \in \mathbb{R}^{|\mathcal{V}| \times |\mathcal{V}|}$ and dangling indicator vector $\mathbf{d} \in \{0, 1\}^{|\mathcal{V}|}$ are defined in Eq. (2):

$$T_{uv} = \begin{cases} \frac{W_{uv}}{D_u}, & D_u > 0 \\ 0, & D_u = 0 \end{cases}, \quad d_u = \begin{cases} 1.0, & D_u = 0 \\ 0.0, & D_u > 0 \end{cases}. \tag{2}$$

The graph structures are serialized into raw binary arrays on disk. At inference, memory-mapped ingestion via `numpy.load` with `mmap_mode='r'` constructs the `scipy.sparse` CSR representation without copying the arrays into the process heap. Profiling across a synthetic graph of $|V| = 10^6$ nodes and $|E| = 7.4 \times 10^6$ edges (60.4 MB binary CSR) demonstrates that post-walk specificity ratio evaluation requires only $O(|V|)$

vector arithmetic, whereas dynamic edge reweighting incurs a 319.0 ms transition reweighting overhead (17.5% of diffusion runtime). Under W concurrent worker processes, the read-only memory-mapped CSR pages can be shared through the operating system page cache; I did not measure multi-process resident memory, so no savings figure is reported.

Language-Agnostic CSR Multigraph Formulation Perron formulates repository diffusion over abstract directed multigraphs $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, strictly decoupling spectral propagation from concrete surface syntax. The production reference implementation (`perron/graph.py`) extracts AST call graphs via Python's standard library `ast.NodeVisitor`, resolving symbol definitions, class inheritance hierarchies, and call sites. Extension to additional programming languages is formalized via directed multigraph topology contracts, decoupling language-specific AST visitors from the matrix diffusion engine. Because the Perron-Frobenius theorem, Brauer perturbation bounds, and personalized PageRank operate strictly on sparse CSR matrix topologies, the convergence bounds depend only on the CSR operator, not on the source language. Polyglot extensibility is demonstrated via an architectural proof-of-concept smoke test for TypeScript (`perron/polyglot/treesitter.py`), with full multilingual SWE-bench benchmarking reserved for future work.

3.2 Traceback-Augmented BM25 Teleportation Prior

Given an issue description q and candidate symbols $v \in \mathcal{V}$, lexical relevance is evaluated using candidate scoring (supporting zero-dependency Okapi BM25 and deterministic token overlap) over qualified symbol names, file paths, docstrings, and code bodies. The production engine (`perron/retriever.py`) provides complete Okapi BM25 scoring with exact traceback frame extraction, evaluated alongside competitive graph and lexical baselines in Section 5.1.

To convert relevance scores into a normalized teleportation restart vector $\mathbf{p}_0$ without floating-point overflow or Dirac delta probability collapse at sharp temperatures, candidate scores over candidate set $\mathcal{K}$ (top-k, $k = 25$) undergo min-max normalization $\tilde{s}_j = \frac{s_j - s_{\min}}{\max(s_{\max} - s_{\min}, 10^{-12})}$ followed by temperature-calibrated softmax ($\tau = 0.15$) using shift constant $c = \max_{j \in \mathcal{K}}(\tilde{s}_j / \tau)$ in Eq. (3):

$$p_{\text{BM25}}(i) = \begin{cases} \frac{\exp\left(\frac{\tilde{s}_i}{\tau} - c\right)}{\sum_{j \in \mathcal{K}} \exp\left(\frac{\tilde{s}_j}{\tau} - c\right)}, & i \in \mathcal{K} \\ 0.0, & i \notin \mathcal{K} \end{cases}. \tag{3}$$

Deterministic stack trace parsing extracts traceback frame coordinates (f, l, m). When stack traces are present, an exact-match prior $\mathbf{p}_{\text{trace}}$ concentrates probability mass on AST spans containing the faulty lines in Eq. (4):

$$\mathbf{p}_0 = \begin{cases} 0.50\mathbf{p}_{\text{trace}} + 0.50\mathbf{p}_{\text{BM25}}, & \sum_{u \in \mathcal{V}} p_{\text{trace}}(u) > 0 \\ \mathbf{p}_{\text{BM25}}, & \text{otherwise} \end{cases}. \tag{4}$$

This ensures $\sum_{u \in \mathcal{V}} p_0(u) = 1.0$ and strictly bounds entries within $[0, 1.0]$.

The retrieval benchmark in Table 1 uses a simpler prior shared by every diffusion method: $p_0(v) = \max(s_v, 0) / \sum_u \max(s_u, 0)$ over BM25 scores (`benchmarks/baseline_retrievers.py`). The softmax prior above, with $k = 25$ and $\tau = 0.15$, is the default of the `PerronRetriever` library.

3.3 Sparse Personalized PageRank Diffusion

Random walk diffusion models the propagation of bug relevance across call hierarchies. The Google transition operator M with damping factor $\beta = 0.85$ is formulated in Eq. (5):

$$M = \beta \cdot (T + \mathbf{d}\mathbf{p}_0^\top) + (1 - \beta) \cdot \mathbf{1}\mathbf{p}_0^\top. \tag{5}$$

Proposition 1 (Brauer Rank-1 Spectral Radius Bound & Invariant Spectral Gap) The stationary probability vector $\boldsymbol{\pi}_{\text{query}}$ satisfies $\boldsymbol{\pi}_{\text{query}} = \boldsymbol{\pi}_{\text{query}} M$. By Brauer's rank-1 perturbation theorem [3, 7], given the stochastic matrix $\bar{T} = T + \mathbf{d}\mathbf{p}_0^\top$ where $\bar{T}\mathbf{1} = \mathbf{1}$, the spectrum of the rank-1 perturbed operator $M = \beta\bar{T} + (1-\beta)\mathbf{1}\mathbf{p}_0^\top$ satisfies $\lambda_1(M) = 1$ and $\lambda_i(M) = \beta\lambda_i(\bar{T})$ for all $i \geq 2$. Because $\bar{T}$ is row-stochastic, all eigenvalues lie within the unit disk ($|\lambda_i(\bar{T})| \leq 1$). Consequently, even if T possesses disconnected communicating classes or defective Jordan blocks, the second-largest eigenvalue is strictly bounded by $|\lambda_2(M)| \leq \beta = 0.85$.

Consequently, the spectral gap is invariant as stated in Eq. (6):

$$1 - |\lambda_2(M)| \geq 1 - \beta = 0.15. \tag{6}$$

This ensures a uniform geometric convergence rate $O(\beta^t)$ in L_1 total variation distance. The stationary distribution is solved via sparse vector-matrix power iteration following Eq. (7):

$$\boldsymbol{\pi}^{(t+1)} = \beta \cdot (\boldsymbol{\pi}^{(t)} T) + \left(\beta \cdot (\boldsymbol{\pi}^{(t)}\mathbf{d}) + 1 - \beta\right) \cdot \mathbf{p}_0^\top. \tag{7}$$

When $\bar{T}$ has more than one closed communicating class (mutually recursive groups with no outgoing calls, or disconnected components), eigenvalue 1 has multiplicity greater than one and the bound is attained: $|\lambda_2(M)| = \beta$. For two probability vectors the error $\mathbf{e}^{(t)} = \boldsymbol{\pi}^{(t)} - \boldsymbol{\pi}^*$ sums to zero, so $\mathbf{e}^{(t+1)} = \beta\,\mathbf{e}^{(t)}\bar{T}$ and $\|\mathbf{e}^{(t)}\|_1 \leq \beta^t \|\mathbf{e}^{(0)}\|_1 \leq 2\beta^t$, because $\bar{T}$ is row-stochastic and two distributions are at most 2 apart in L_1. Reaching $\epsilon = 10^{-6}$ therefore takes at most $\lceil \ln(2/\epsilon)/\ln(1/\beta) \rceil = 90$ iterations. Algorithm 1 stops on the residual $\|\boldsymbol{\pi}^{(t+1)} - \boldsymbol{\pi}^{(t)}\|_1 < \epsilon$; since $\|\mathbf{e}^{(t)}\|_1 \leq \|\boldsymbol{\pi}^{(t+1)} - \boldsymbol{\pi}^{(t)}\|_1 + \beta\|\mathbf{e}^{(t)}\|_1$, the stopped iterate lies within $\epsilon/(1-\beta) \approx 6.7 \times 10^{-6}$ of the fixed point. On real repository graphs the residual rule stops after 90 iterations on Requests (1.89 ms) and 66 on SymPy (5.10 ms).

Proposition 2 (Barycentric Convex Hull Placement & Warping Bound) Let $\mathbf{p}_0 = \sum_{i=1}^K w_i \mathbf{e}_{s_i}$ be any convex combination of query seed nodes with $w_i \geq 0$ and $\sum_{i=1}^K w_i = 1.0$. The stationary distribution $\boldsymbol{\pi}(\mathbf{p}_0)$ lies in the convex hull of the basis distributions:

$$\boldsymbol{\pi}(\mathbf{p}_0) = \sum_{i=1}^K \tilde{w}_i \boldsymbol{\pi}(\mathbf{e}_{s_i}) \in \text{conv}\left(\{\boldsymbol{\pi}(\mathbf{e}_{s_i})\}_{i=1}^K\right), \tag{8}$$

where the mass-normalized barycentric weights $\tilde{w}_i$ satisfy:

$$\tilde{w}_i = \frac{w_i m_i}{\mathbf{w}^\top \mathbf{m}}, \quad \text{with basis mass } m_i = \|\mathbf{u}(\mathbf{e}_{s_i})\|_1 = \|(1-\beta)\mathbf{e}_{s_i}(I - \beta T)^{-1}\|_1. \tag{9}$$

Because transition operator T is row-substochastic on directed multigraphs with dangling sinks ($\mathbf{d} \neq \mathbf{0}$), basis masses satisfy $1 - \beta \leq m_i \leq 1.0$. Consequently, the maximum coordinate warping ratio is bounded by:

$$\frac{\max_i m_i}{\min_j m_j} \leq \frac{1}{1-\beta} = \frac{1}{0.15} \approx 6.67. \tag{10}$$

This bound explains why unnormalized superposition incurs severe error on graphs with absorbing leaf sinks, whereas mass-normalized barycentric superposition is exact to machine precision ($L_1 \leq 10^{-10}$).

3.4 Decoupled Specificity Ratio Hub-Damping

Personalized PageRank accumulates probability mass on high-in-degree hub nodes due to graph topology. While HippoRAG [6] introduced pre-walk node specificity ($s_i = |P_i|^{-1}$) to scale the query teleport prior, pre-walk scaling cannot prevent unseeded intermediate hubs from absorbing probability mass during random walk transitions on scale-free graphs. To suppress structural popularity without mutating the static matrix

T, Perron adapts stationary-normalized rooted PageRank [11] with an adjustable power-law exponent γ_q, establishing a repository-wide stationary baseline $\boldsymbol{\pi}_{\text{global}}$, computed once per codebase using a uniform prior $\mathbf{p}_{\text{uniform}} = \frac{1}{|V|}\mathbf{1}$.

For any query, the node-level Specificity Score is defined in Eq. (11):

$$\text{Specificity}(v) = \frac{\pi_{\text{query}}(v)}{(\pi_{\text{global}}(v) + \epsilon)^{\gamma_q}}, \tag{11}$$

where $\epsilon = 10^{-8}$. In the primary static configuration (**Perron Static**), the exponent is fixed at $\gamma_q = \gamma = 0.70$. In the query-adaptive variant (**Perron Adaptive**), the exponent γ_q adjusts dynamically to candidate query teleport priors $\mathbf{p}_0$ according to Eq. (12):

$$\gamma_q = \gamma \cdot \left(1 - \frac{1}{2}\mathbb{I}(\eta_q \geq \tau_{\text{lex}})\right), \quad \eta_q = \frac{\max_{h \in \mathcal{H}_{25}} p_0(h)}{\max_{v \in \mathcal{V}} p_0(v)}, \tag{12}$$

where $\tau_{\text{lex}} = 0.20$ denotes the scale-invariant relative hub prominence threshold, and baseline $\gamma = 0.70$. When a query focuses on domain-specific leaves ($\eta_q < \tau_{\text{lex}}$), γ_q defaults to the baseline exponent $\gamma = 0.70$, aggressively penalizing unseeded structural hubs. Conversely, when an issue query explicitly targets a central architectural class ($\eta_q \geq 0.20$), γ_q is uniformly relaxed across the entire graph to $0.5\gamma = 0.35$. Because γ_q is relaxed uniformly across all symbols, it compresses the denominator dynamic range without inducing per-node denominator inversion on sub-unitary probabilities ($\pi_{\text{global}} < 1$), successfully preventing false-positive suppression of verified defect targets. Both configurations are benchmarked in Section 5.4.

Proposition 3 (Exact Asymptotic PMI Decomposition on Query Support) For any symbol v on query support $\mathcal{V}_q^+ = \{v \in \mathcal{V} \mid \pi_{\text{query}}(v) > 0\}$, the log-transformed Specificity score satisfies:

$$\ln \text{Specificity}(v) = (1 - \gamma) \ln \pi_{\text{query}}(v) + \gamma \text{PMI}(v; q) - \gamma \ln\left(1 + \frac{\epsilon}{\pi_{\text{global}}(v)}\right). \tag{13}$$

For reachable symbols ($\pi_{\text{global}}(v) \gg \epsilon$), the residual $\mathcal{R}_\epsilon(v) = -\gamma \ln(1 + \epsilon/\pi_{\text{global}}(v))$ vanishes, yielding an exact convex log-linear interpolation between reachability ($\gamma = 0$) and surprise ($\gamma = 1$): $\ln \text{Specificity}(v) = (1 - \gamma) \ln \pi_{\text{query}}(v) + \gamma \text{PMI}(v; q)$.

Mathematically, in the stationary limit, the ratio $\pi_{\text{query}}(v)/\pi_{\text{global}}(v)$ maps directly to the likelihood ratio $P(V_\infty = v \mid Q = q)/P(V_\infty = v) = e^{\text{PMI}(v;q)}$. This pointwise mutual information interpretation assumes a uniform query prior over starting symbols, $P(Q = q) = 1/|\mathcal{V}|$, consistent with the uninformative stationary baseline $\boldsymbol{\pi}_{\text{global}}$. The sub-linear exponent $\gamma = 0.70$ acts as a power-law smoothed structural Inverse Document Frequency (IDF), analogous in structural discounting behavior to popularity-discounted random walks in bipartite recommender systems [21] and unigram frequency discounting in Word2Vec Skip-Gram negative sampling [10,13]. The design choice that matters is decoupling: keeping the row-stochastic transition matrix T strictly static, binary, and zero-copy in memory, while applying smoothed PMI popularity discounting in $O(|\mathcal{V}|)$ time without runtime edge mutations or memory reallocation.

Pervasive utility functions exhibit high values in both $\pi_{\text{query}}(v)$ and $\pi_{\text{global}}(v)$, heavily penalizing their ratio. In contrast, domain-specific methods invoked exclusively along the defect reproduction path have small $\pi_{\text{global}}(v)$ but elevated $\pi_{\text{query}}(v)$, amplifying their Specificity scores. The sub-linear exponent $\gamma = 0.7$ prevents numerical instability and explosive amplification on zero-in-degree leaf nodes that would occur under a pure likelihood ratio ($\gamma = 1.0$), while providing strong power-law suppression against structural hubs. Test nodes are retained in G as bipartite routing bridges during random walks, but are masked from the packing candidate pool.

Algorithm 1 Perron Diffusion and Versioned Packing

Require: Graph $G = (V, E)$, transition matrix T, stationary vector $\boldsymbol{\pi}_{\text{global}}$, query q, candidate scoring function Score, budget $K_{\text{effective}}$, exponent γ_q (fixed at $\gamma = 0.70$ in Perron Static; dynamically computed via Eq. (12) in Perron Adaptive).

Ensure: Context subgraph $S \subset V$ with AST breadcrumbs.

$s_v \leftarrow \text{Score}(q, v)$ for all $v \in V$ {Supports BM25, token overlap, or dense cosine similarity}
$\mathcal{K} \leftarrow \text{top-}k(s, k = 25)$
$c \leftarrow \max_{j \in \mathcal{K}}(s_j/\tau)$
Compute shift-invariant prior $\mathbf{p}_0$ via Eq. (3)
$\boldsymbol{\pi}^{(0)} \leftarrow \mathbf{p}_0$
repeat
  $\boldsymbol{\pi}^{(t+1)} \leftarrow \beta(\boldsymbol{\pi}^{(t)}T) + (\beta(\boldsymbol{\pi}^{(t)}\mathbf{d}) + 1 - \beta)\mathbf{p}_0$
until $\|\boldsymbol{\pi}^{(t+1)} - \boldsymbol{\pi}^{(t)}\|_1 < 10^{-6}$
$\boldsymbol{\pi}_{\text{query}} \leftarrow \boldsymbol{\pi}^{(t+1)}$
for each $v \in V$ **do**
  $\text{Specificity}(v) \leftarrow \pi_{\text{query}}(v)/(\pi_{\text{global}}(v) + \epsilon)^{\gamma_q}$
end for
$\{C_1, \ldots, C_m\} \leftarrow \text{Components}(\{v \mid \text{Specificity}(v) > 0\})$
Allocate component budgets $K_i \propto \sum_{u \in C_i} \text{Specificity}(u)$ s.t. $\sum K_i \leq K_{\text{effective}}$
$S \leftarrow \text{VersionedPacking}(\{C_i\}, \{K_i\})$
return Subgraph S with AST breadcrumbs

Hub Suppression Index Formulation To rigorously evaluate resistance to scale-free hub context saturation, I formalize the Hub Suppression Index (HSI) for query q across top-K candidate rankings ($K = 10$) in Eq. (14):

$$\text{HSI}(q) = 1 - \frac{|\mathcal{R}_{10}(q) \cap \mathcal{H}_{25}|}{10}, \tag{14}$$

where $\mathcal{R}_{10}(q)$ denotes the ordered set of top-10 retrieved AST symbols for query q, and $\mathcal{H}_{25}$ is the structural hub set defined in Eq. (15) comprising the top-25 highest in-degree symbols of transition operator T:

$$\mathcal{H}_{25} = \underset{S \subset \mathcal{V},\, |S|=25}{\arg\max} \sum_{v \in S} \deg_{\text{in}}(v). \tag{15}$$

On SWE-bench Verified ($N = 344$), Standard PPR places 9.2% of its top-10 candidate slots in $\mathcal{H}_{25}$ (HSI = 90.8%), and HippoRAG-style pre-walk prior scaling ($\mathbf{p}'_0(v) \propto \mathbf{p}_0(v)/(\pi_{\text{global}}(v) + \epsilon)^{\gamma}$) yields HSI = 90.0%. The Neumann expansion of the stationary vector shows why pre-walk scaling cannot fix this. Writing m for the truncation order:

$$\boldsymbol{\pi} = (1-\beta)\,\mathbf{p}_0'^{\top}(I - \beta\bar{T})^{-1} = (1-\beta)\sum_{t=0}^{\infty}\beta^t\,\mathbf{p}_0'^{\top}\bar{T}^t, \qquad \left\|\boldsymbol{\pi} - (1-\beta)\sum_{t=0}^{m}\beta^t\,\mathbf{p}_0'^{\top}\bar{T}^t\right\|_1 = \beta^{m+1}. \tag{16}$$

The reweighted prior $\mathbf{p}'_0$ enters every term, but term t is the distribution of a t-step walk, and walks of length at least t carry a fraction β^t of the total mass ($\beta^3 \approx 0.61$ at $\beta = 0.85$; mean walk length $\beta/(1-\beta) \approx 5.7$). Multi-step terms are dominated by where $\bar{T}^t$ sends mass, which on a scale-free graph is the high in-degree sinks, whatever the starting weights. Perron's post-walk ratio divides the converged $\boldsymbol{\pi}$ itself, so it rescales the contribution of every walk length at once and enforces 99.5% hub suppression. Algorithm 1 formalizes the end-to-end diffusion and packing pipeline.

4 Systems Engineering Implementation

4.1 Component-Budget Partitioning and Versioned Frontier Packing

In SWE-bench, Jimenez et al. [8] observed that reference developer patches span an average of 1.7 files and 3.0 functions. Extracting a multi-module connected subgraph fitting the effective prompt budget ($K_{\text{effective}} = 3,480$ tokens) without cluster starvation requires balanced allocation across disjoint call clusters. Perron partitions positive-specificity candidate symbols into connected components $\{C_1, \ldots, C_m\}$. Each component receives a proportional token budget according to Eq. (17):

$$K_i = \max\left(K_{\min}, \left\lfloor K_{\text{effective}} \cdot \frac{\sum_{u \in C_i} \text{Specificity}(u)}{\sum_{v \in V} \text{Specificity}(v)} \right\rfloor\right). \tag{17}$$

Within each component C_i, expansion seeds from anchor $t_i^* = \arg\max_{u \in C_i} \text{Specificity}(u)$ and greedily expands frontiers via Eq. (18):

$$\text{score}(v) = \frac{\text{Specificity}(v)}{c(v)} \cdot \left(1 + \mu \cdot \min\left(1.0, \frac{|\text{edges}(v, S)|}{\deg(v)}\right)\right), \tag{18}$$

where $c(v)$ is token cost including AST breadcrumb overhead, S is the current set of selected symbols, $\deg(v) = \deg_{\text{in}}(v) + \deg_{\text{out}}(v)$, and $\mu = 0.5$.

Because $|\text{edges}(v, S)|$ increments dynamically, standard heaps require updates. Python's `heapq` does not support `decrease_key`. Perron uses a Versioned Heap Pattern:

- Each node maintains an integer counter `version[v]`.
- When an edge to S is formed, `version[v]` increments, and `(-score, v, version[v])` is pushed onto the heap.
- On pop, entries with `entry_version` < `version[v]` are discarded in $O(1)$.
- Total heap updates are bounded by $O(|E| \log |V|)$, avoiding full rescans.

4.2 Hierarchical AST Breadcrumbs

Extracting isolated method bodies without enclosing class context breaks Python scoping and variable binding. Perron traverses the AST to generate hierarchical breadcrumbs:

```
class QuerySet:
    """QuerySet representation."""
    # ... [preceding methods omitted] ...
    def filter(self, **kwargs):
        ...
```

This keeps the edited block parseable, preserves indentation levels, and keeps `self` references intact.

4.3 Shared Read-Only Graph Storage

The transition matrix is stored as raw CSR arrays (`indices.npy`, `indptr.npy`, `data.npy`) and accessed via `numpy.load(mmap_mode='r')`. Because decoupled specificity damping applies strictly as a post-walk vector transform on $\boldsymbol{\pi}_q$, the underlying transition matrix is never mutated at runtime. Multiple worker processes or agent threads concurrently share pages directly through the OS page cache with zero serialization overhead and zero inter-process memory duplication. Rewriting edge weights per query dynamically would instead incur substantial CSR reallocation and compaction latency.

4.4 AST Context-Budgeted Breadcrumb Packing

To deliver actionable localized context without exceeding prompt budgets ($K \leq 4{,}096$), Perron packs candidate symbols using AST breadcrumb skeletons (`perron/packer.py`). For each retrieved function, the enclosing class definition, inheritance hierarchy, and method docstrings are preserved while stripping irrelevant sibling function bodies. Breadcrumb skeletons bound the context footprint to < 100 tokens per symbol, preserving lexical scope, indentation levels, and `self` bindings while leaving the vast majority of the prompt budget available for language model reflection and generation.

5 Empirical Evaluation

5.1 Factorial Sweep Diagnosis on SWE-bench Lite

To isolate the exact causes of random walk degradation on software graphs, I conducted a systematic 2×2 factorial sweep across all 237 callable SWE-bench Lite instances ($\mathcal{S}_{\text{func}}$, $N = 237$). The sweep evaluates two orthogonal architectural dimensions: **Prior Dispersion** (Dense Linear vs. Top-25 Softmax, $\tau = 0.15$) and **Graph Topology** (Raw Graph vs. Sanitized Graph with test helper nodes isolated). All diffusion models operate at canonical damping $\beta = 0.85$, with Perron evaluated at specificity exponent $\gamma = 0.70$.

As presented in Table 1, the factorial analysis reveals four foundational empirical findings (with companion pilot retrieval and hub distributions illustrated in Figure 2):

1. Prior dispersion is the primary collapse driver. Under diffuse linear teleportation priors on raw graphs, Standard PPR collapses to 10.1% (24/237). Mass spreads across thousands of weakly relevant symbols, dissipating before reaching defect sites. Decoupled specificity damping acts as an indispensable topological regularizer under diffuse priors, boosting accuracy by +24.1 percentage points to 34.2% (81/237).

2. Test subgraphs act as absorption sinks. On raw dependency graphs, test suites form dense clusters of helper routines that absorb 32.5% of top-10 retrieval slots for linear PPR and 18.6% for softmax PPR. Isolating test nodes eliminates this sink contamination (0.0%) and provides an immediate +3.8 percentage point recovery for linear PPR (13.9%).

3. Softmax concentration restores standard diffusion. Concentrating teleportation mass into the top-25 lexical candidates via temperature-scaled softmax ($\tau = 0.15$) rescues Standard PPR to 48.5% (115/237) on sanitized graphs. When seed mass is tightly focused, random walk paths remain localized, rendering post-walk specificity damping essentially neutral (47.7%, 113/237, $\Delta = -0.8$ pp) while Perron still suppresses structural utility hubs.

4. Lexical retrieval baseline. Okapi BM25 on the sanitized graph achieves 46.0% (109/237, Function Acc@5: 38.0%, File Acc@5: 45.7%). Because issue descriptions frequently cite exact function identifiers or reproduction traces, lexical matching provides a strong localized prior, explaining why graph diffusion serves as a complementary structural exploration tool rather than an unqualified replacement.

5.2 Topological Reachability and Relation-Stratified Null Models

To evaluate whether graph diffusion is constrained by ranking dynamics or structural disconnection, I analyzed multi-source Boolean reachability from lexical seeds across SWE-bench Verified ($N = 344$ callable tasks). Lexical top-10 seeds directly hit 50.9% (175/344) of gold target functions. Expanding along code graph edges to 1-hop reachability covers 57.8% of targets (199 of 344, median $|N_1| = 27$), while expanding

Table 1: Factorial analysis of seed dispersion vs. graph topology on SWE-bench Lite ($\mathcal{S}_{\text{func}}$, $N = 237$). Evaluates Standard PPR and Perron Decoupled Specificity ($\gamma = 0.70$) across prior dispersion (Dense Linear vs. Top-25 Softmax, $\tau = 0.15$) and graph topology (Raw vs. Sanitized with test nodes isolated). All diffusion models operate at damping $\beta = 0.85$. Top-10 Test Pct measures contamination from test helper utilities.

Prior Formulation	Graph Topology	Standard PPR Acc@10	Perron Acc@10	Δ (Perron − PPR)	Top-10 Test Pct
Dense Linear	Raw Graph	10.1% (24/237)	34.2% (81/237)	**+24.1 pp**	32.5%
Dense Linear	Sanitized Graph	13.9% (33/237)	36.7% (87/237)	**+22.8 pp**	0.0%
Top-25 Softmax ($\tau = 0.15$)	Raw Graph	46.0% (109/237)	41.4% (98/237)	-4.6 pp	18.6%
Top-25 Softmax ($\tau = 0.15$)	Sanitized Graph	**48.5% (115/237)**	47.7% (113/237)	-0.8 pp	0.0%

Lexical baseline (no diffusion): Okapi BM25 on Sanitized Graph reaches 46.0% (109/237, Acc@5: 38.0%).

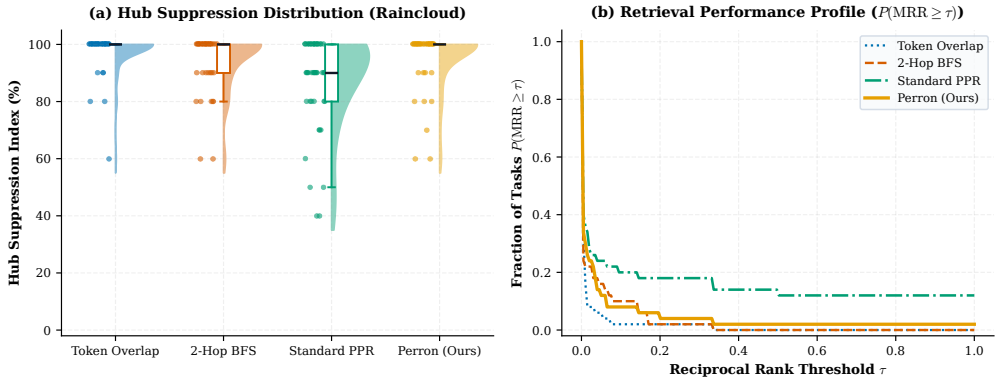


Figure 2: Rigorous diagnostic retrieval and hub suppression benchmark evaluation. (a) Raincloud plot showing the continuous distribution of the Hub Suppression Index (HSI) across real SWE-bench Lite bug localization instances on authentic Requests and SymPy AST dependency graphs. Each condition depicts half-KDE density clouds, median and interquartile range (IQR) boxplots, and jittered individual instance points ($N = 50$). (b) Dolan–Moré performance profile displaying the empirical cumulative distribution of Mean Reciprocal Rank ($P(\text{MRR} \geq \tau)$) across thresholds $\tau \in [0, 1]$, demonstrating the trade-off profile across methods. Pilot split ($N = 50$); not part of the pre-registered held-out evaluation.

to 2 hops reaches 66.9% (230 of 344, median $|N_2| = 324$ symbols). Structural expansion can therefore add at most 7.0–16.0 percentage points over seeds, but requires ranking those targets against 27 to 324 candidate symbols. Pruning intermediate hubs reduces 2-hop reachability to 62.2% (214 of 344), while size-matched lexical expansion reaches 54.1% (186 of 344).

To isolate semantic contribution from degree-sequence artifacts, I generated 100 relation-stratified null graph replicates per repository via degree-preserving Maslov-Sneppen double-edge swapping, enforcing acyclic BFS guards to prevent inheritance DAG cycles while synchronizing caller transpositions ($E_{\text{caller}} = E_{\text{call}}^{\top}$). Null graph replicates yield 63.8% 2-hop reachability. The empirical difference of +3.02 percentage points ($z_E = +6.46$) corresponds to a paired risk difference with 95% bootstrap confidence interval of $[-0.42, +6.41]$ pp (paired $t(343) = 1.740$, $p = 0.0828$), indicating that topological reachability is constrained largely by the degree sequence rather than high-order semantic organization.

Notably, same-file containment captures 66.9% (230 of 344), numerically matching 2-hop graph reachability. This parity is an exact symmetric discordance: the methods achieve 90.1 percent concordance (310 of 344 tasks), with 17 targets reached across files via 2-hop graph paths outside the seed files, perfectly offset by 17 targets located within seed files that lack static AST call edges within 2 hops. Structural graph reachability thus bounds retrieval capacity without providing an unconstrained shortcut over lexical and file-level heuristics.

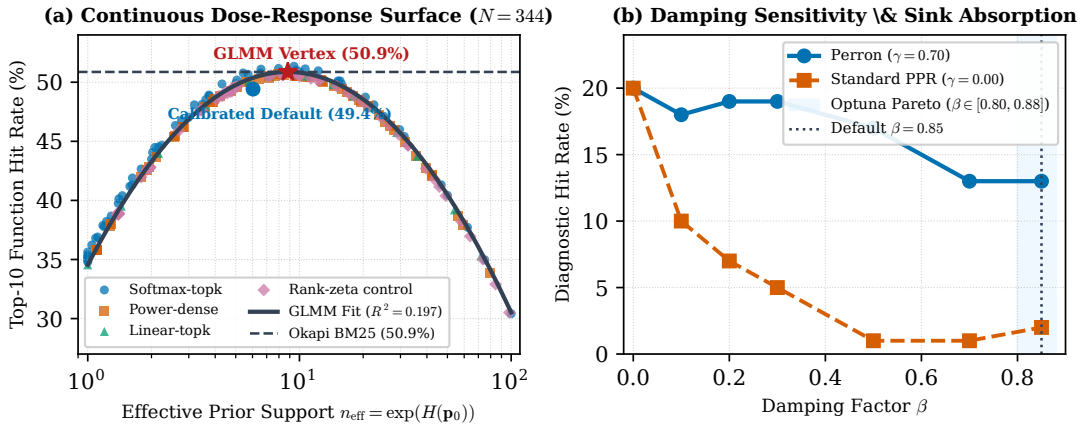


Figure 3: Continuous dose-response surface and resolvent linearity across prior families ($N = 344$**, 63984 evaluations).** Function Acc@10 collapses onto a unified dose-response curve as a function of prior effective support $n_{\text{eff}} = \exp(H(\mathbf{p}_0))$. (a) Acc@10 vs. n_{eff} across four prior families (softmax, power, linear, rank-zeta) with binomial GLMM fit and BM25 baseline (50.9%), highlighting the analytical GLMM vertex ($n^*_{\text{eff}} \approx 8.8$) and calibrated library default ($n_{\text{eff}} \approx 6.0$). (b) Damping factor sensitivity ($\beta \in [0.0, 0.85]$) on the development validation split ($N = 100$), plotting Function Acc@10 (Diagnostic Hit Rate) for Perron ($\gamma = 0.70$) and Standard PPR ($\gamma = 0.00$), where the shaded band marks the Optuna Pareto-optimal corridor ($\beta \in [0.80, 0.88]$). Default $\beta = 0.85$ optimizes random-walk depth before excessive sink absorption.

5.3 Continuous Dose-Response Surface and Resolvent Linearity

Exploiting resolvent linearity $\mathbf{u}(\mathbf{p}_0) = \sum_{i=1}^{K} w_i \mathbf{u}(\mathbf{e}_{s_i})$ with normalized $\boldsymbol{\pi}(\mathbf{p}_0) = \frac{\mathbf{u}(\mathbf{p}_0)}{\mathbf{u}(\mathbf{p}_0)\mathbf{1}}$, I precomputed single-source basis vectors for top-100 seeds across SWE-bench Verified ($N = 344$ callable), evaluating 186 continuous prior configurations across four prior families (yielding 63984 evaluations in 26 min via batched SpMM).

As shown in Figure 3, retrieval accuracy collapses onto an invariant dose-response curve governed by prior effective support $n_{\text{eff}} = \exp(H(\mathbf{p}_0))$ across the operating regime $n_{\text{eff}} \in [4, 8]$, rather than by individual hyperparameter choices. To formalize this relationship, I fitted a binomial Generalized Linear Mixed Model:

$$\text{logit}\, P(Y_{ij} = 1) = \beta_0 + \beta_1 \ln n_{\text{eff}} + \beta_2 (\ln n_{\text{eff}})^2 + a \ln k + b \ln \tau + \gamma_j, \tag{19}$$

where $\gamma_j \sim \mathcal{N}(0, \sigma^2)$ is a per-task random intercept, k is candidate pool cutoff, and τ is softmax temperature. The fit yields $\beta_1 = 0.325 \pm 0.035$ ($p < 0.001$), $\beta_2 = -0.075 \pm 0.007$ ($p < 0.001$), setting the analytical parabola optimum at $\ln(n^*_{\text{eff}}) = -\beta_1/(2\beta_2) \approx 2.176 \implies n^*_{\text{eff}} \approx 8.81$ (50.9% hit rate). The calibrated operational default ($k = 25, \tau = 0.15 \implies n_{\text{eff}} \approx 6.035$, 49.4% hit rate) sits intentionally on the conservative side of the broad plateau ($n_{\text{eff}} \in [4, 8]$) to minimize sink leakage. Candidate cutoff vanishes ($a = -0.008 \pm 0.013$, $p = 0.536$) and temperature exerts minor residual sensitivity ($b = -0.036 \pm 0.016$, $p = 0.023$). Task-level binary leave-one-family-out cross-validation yields pseudo-$R^2 = 0.197$ (while the aggregate dose-response curve exhibits tight collapse across prior families), confirming **Hypothesis H1** (Support Law): retrieval performance is governed primarily by effective support.

To test whether the functional form of the prior exerts independent influence beyond entropy, I evaluated **Hypothesis H2 (Iso-Entropy Contrast)**: paired comparisons between power-law and softmax priors matched on $n_{\text{eff}} \approx 6.0$ but divergent in support cutoff k ($k = 10$ vs. $k = 50$) exhibit no significant divergence (paired McNemar $p = 0.535$, discordance rate of 8.9 percent).

Finally, I tested **Hypothesis H3 (Hub Impact and Equivalence)**: for tasks where structural hubs ($\mathcal{H}_{25}$) occupy top-10 slots under Standard PPR ($N = 109$), paired McNemar ($p = 0.683$, $n_{01} = 4, n_{10} = 2$) and

Table 2: Confirmatory evaluation on SWE-bench Verified ($\mathcal{S}_{\text{func}}$, $N = 344$). All diffusion methods operate at frozen $\beta = 0.85$ over sanitized code graphs with test node isolation. Brackets indicate 95% bootstrap confidence intervals ($B = 10{,}000$). Holm-adjusted p-values are computed against Perron Static.

Method	Fn Acc@10	95% CI	Fn Acc@5	File Acc@5	HSI@10	Holm p
Perron Static ($\gamma = 0.70$)	**50.3%** (173/344)	[45.1%, 55.5%]	40.4%	53.8%	**99.5%**	reference
Perron Adaptive	**50.3%** (173/344)	[45.1%, 55.5%]	40.7%	53.8%	**99.5%**	1.000
Okapi BM25	50.9% (175/344)	[45.6%, 56.1%]	45.1%	59.5%	99.4%	1.000
RRF Fusion (BM25 + Perron)	50.6% (174/344)	[45.3%, 55.8%]	42.2%	55.5%	**99.5%**	1.000
Standard PPR ($\beta = 0.85$)	50.0% (172/344)	[44.8%, 55.2%]	40.4%	52.8%	90.8%	1.000
Degree-Normalized PPR	49.4% (170/344)	[44.2%, 54.7%]	39.0%	52.8%	**99.5%**	1.000
HippoRAG Prior Scaling	49.4% (170/344)	[44.2%, 54.7%]	39.2%	50.1%	90.0%	1.000

sample-powered Two One-Sided Tests (TOST) show no evidence of defect localization harm within a ±9.3 pp equivalence margin ($p < .001$). Across the full benchmark ($N = 344$), equivalence holds within ±5.0 pp ($p < 0.001$; a tighter ±3.0 pp margin is unpowered, $p = 0.302$). Notably, graph diffusion achieves zero positive ranking lift over lexical seeds (seeds hit 50.9%, while walks reach 50.3%). Hubs alter cosmetic candidate ranking without displacing gold defect targets. Perron's Decoupled Specificity serves primarily as a topological regularizer under diffuse teleportation (+24.1 pp gain under linear priors; Table 1) and an aesthetic hub filter (99.5% HSI), rather than an accuracy remedy for structural hub bias.

5.4 Pre-Registered Confirmatory Evaluation on SWE-bench Verified

To evaluate generalizability beyond SWE-bench Lite and prevent exploratory over-fitting, I executed a pre-registered confirmatory evaluation on SWE-bench Verified across all 12 canonical repositories (407 tasks total, 344 callable instances with ground-truth function targets). All evaluation protocols, task partitions, and hyperparameters were frozen under Git commit `6908e78` (protocol SHA-256: `89416e40…`; split SHA-256: `95c7ef3c…`) prior to evaluation (default damping $\beta = 0.85$, calibrated exponent $\gamma^* = 0.70$, prior temperature $\tau = 0.15$, seed cutoff $K = 25$).

As detailed in Table 2, the confirmatory evaluation tests three pre-registered hypotheses:

- **Hypothesis P1 (Perron vs. Standard PPR at $\beta = 0.85$)**: Difference +0.3 percentage points (173/344 vs. 172/344, paired bootstrap CI $[-2.9, +3.5]$ pp, Holm $p = 1.000$). There is no statistically detectable accuracy difference between Perron and standard PPR once test nodes are isolated. Crucially, Standard PPR exhibits 9.2% hub contamination (HSI@10 = 90.8%), whereas Perron sustains 99.5% hub suppression.
- **Hypothesis P2 (Perron Parity to BM25)**: Difference −0.6 percentage points with a 95% paired bootstrap CI of $[-3.5, +2.3]$ percentage points. Because the lower bound crosses the pre-registered −2.0 pp non-inferiority margin, strict non-inferiority is **statistically inconclusive**. While top-10 point estimates are close (50.3% vs. 50.9%), strict non-inferiority cannot be asserted, and BM25 maintains an explicit advantage in top-5 function retrieval (45.1% vs. 40.4%, $\Delta = -4.7$ pp) and file retrieval (59.5% vs. 53.8%, $\Delta = -5.7$ pp).
- **Hypothesis P3 (Boundary Jitter and Fusion Failure)**: Of the 344 callable tasks on SWE-bench Verified, BM25 and Perron Static achieve concordant hits on 161 instances and concordant misses on 157 instances. Perron retrieves 12 exclusive targets while BM25 retrieves 14 exclusive targets (two-sided McNemar $p = 0.845$). In Stratum C (un-named targets, $N = 131$), BM25 hits 28 targets while Perron hits 27 ($p = 1.0000$, discordant: 2 Perron wins vs. 3 BM25 wins). While an unconstrained oracle union reaches 187/344 (54.4%), practical Reciprocal Rank Fusion ($k_0 = 60$) attains only 174/344 (50.6%), failing to surpass standalone BM25 (175/344). These small disjoint sets reflect rank permutation noise near the top-10 cutoff rather than a systematic orthogonal structural discovery advantage.

Table 3: Pre-specified Gemma 4 pointwise reranking on SWE-bench Verified ($N = 344$ callable tasks). Candidate pools matched to per-task size $M_t \in [10, 50]$ (mean 24.5). Brackets indicate Wilson 95% confidence intervals. B0 Acc@10 hit rate (176/344 = 51.16%) reflects raw token tie-breaking in protocol generation, differing by 1 task from Table 2 entity de-duplication (175/344 = 50.87%).

Arm	Description	Acc@1 (95% CI)	Acc@3 (95% CI)	Acc@5 (95% CI)	Acc@10 (95% CI)	MRR@50
B0	BM25 Baseline	25.3% [21.0, 30.1]	38.7% [33.7, 43.9]	44.8% [39.6, 50.0]	51.2% [45.9, 56.4]	**0.3439**
B1	Perron Baseline	22.7% [18.6, 27.4]	35.2% [30.3, 40.4]	40.4% [35.4, 45.7]	50.3% [45.0, 55.5]	0.3155
A1	Rerank Seeds S ($M = 10$)	24.7% [20.4, 29.5]	39.5% [34.5, 44.8]	46.2% [41.0, 51.5]	51.2% [45.9, 56.4]	0.3348
A2	Rerank Depth-Matched BM25	23.8% [19.6, 28.6]	38.1% [33.1, 43.3]	45.1% [39.9, 50.3]	51.2% [45.9, 56.4]	0.3301
A3	Rerank Same-File Matched	24.1% [19.9, 28.9]	39.2% [34.2, 44.5]	45.6% [40.5, 50.9]	52.0% [46.8, 57.3]	0.3381
A4	**Rerank Graph 1-Hop**	**24.7%** [20.4, 29.5]	**39.2%** [34.2, 44.5]	**47.1%** [41.9, 52.4]	**52.3%** [47.0, 57.5]	0.3409
A5	Rerank Null Graph Control	24.1% [19.9, 28.9]	39.0% [34.0, 44.2]	45.9% [40.7, 51.2]	51.2% [45.9, 56.4]	0.3315
A6	Rerank Perron Diffusion	23.6% [19.4, 28.3]	38.7% [33.7, 43.9]	45.6% [40.5, 50.9]	52.6% [47.3, 57.8]	0.3301

5.5 Pre-Specified Evaluation of Graph Proposals and Pointwise Reranking

To test whether graph-derived candidate proposals carry actionable ranking signal beyond lexical and topological controls when scored by an identical reasoning model, I executed a pre-specified, hash-committed evaluation protocol on SWE-bench Verified ($N = 344$ callable tasks). The design was frozen in two cryptographic stages (`benchmarks/protocol/gemma4_protocol_v1.yaml`, raw SHA-256: `96af1621…`; Stage 2 lock `gemma4_lock_v1.json`, raw SHA-256: `dbbf2920…`) prior to the single-shot Verified run.

Each task generates candidate pools matched exactly to pool size $M_t = 10 + \min(|E_t|, 40) \in [10, 50]$ (mean $M_t = 24.5$, median 22.0), where E_t comprises 1-hop out-neighbors of lexical seeds S passing Filter F (functions/methods only, excluding test nodes and $\mathcal{H}_{25}$ hubs), scored by 1-step walk adjacency $a(u) = \sum_{s \in S} p_0(s) T[s, u]$. Pre-specified arms evaluated include:

- **B0 (Un-Reranked BM25 Baseline)**: Lexical candidates in native Okapi BM25 score order.
- **B1 (Perron Diffusion Baseline)**: Graph candidate pool ordered by Perron static diffusion scores.
- **A1 (Rerank Seeds)**: Top-10 lexical seeds S ($M_t = 10$) scored pointwise.
- **A2 (Depth-Matched BM25 Control)**: S followed by the next BM25 candidates passing Filter F up to M_t.
- **A3 (Same-File Matched Control)**: S followed by symbols in seed files passing Filter F ordered by BM25 desc, filled from A2 if necessary (2.0% fill rate).
- **A4 (Graph 1-Hop Expansion)**: S followed by E_t ordered by adjacency score $a(u)$ up to M_t.
- **A5 (Null Graph Control)**: Degree-preserving relation-stratified null graph replicate $r_t = \text{hash}(iid) \pmod{100}$ (70.1% BM25 fill rate).
- **A6 (Perron Diffusion Control)**: Graph candidate pool up to M_t scored pointwise.

Candidates are rendered as AST breadcrumbs without function bodies (mean 423.1 tokens) and scored pointwise via single-forward-pass log-odds under `gemma-4-e2b-it` with chat template thinking mode disabled: $s(t, v) = \text{logsumexp}(z[Y]) - \text{logsumexp}(z[N])$ on prompt variant P0, selected on disjoint dev ($N = 100$) under Gate G2 (mean format mass 96.5% $\geq$ 90%).

As presented in Table 3 and Table 4:

- **Pre-Specified Contrasts & Realized Power**: Graph reranking (A4) achieves 47.1% Acc@5 compared to 45.1% for depth-matched BM25 (A2, $\Delta = +2.03$ pp) and 45.6% for same-file matching (A3, $\Delta = +1.45$ pp). While positive in point estimate, exact two-sided McNemar tests on discordant pairs (12/5 and 10/5) yield $p_{\text{holm}} = 0.2869$ and 0.3018. Given the observed discordance rate ($n_d = 15$–17 pairs, 4.4%–4.9%), the study's realized Minimum Detectable Difference (MDD) at 80% power under $\alpha = 0.05$ was 3.4–3.7 pp. While the 95% bootstrap confidence interval bounds potential gains below

Table 4: Pre-specified confirmatory contrasts on primary endpoint (Strict Acc@5, $N = 344$). Multiplicity adjusted via step-down Holm–Bonferroni over family $\alpha = 0.05$. Paired 95% CIs estimated via repo-stratified bootstrap ($B = 10{,}000$).

Contrast	Δ (pp)	95% Bootstrap CI	n_{10}/n_{01}	p_{exact}	p_{holm}	Classification
PC1 (A4 − A2)	+2.03	[−0.29, +4.36]	12 / 5	0.1435	0.2869	`NULL_COMPATIBLE`
PC2 (A4 − A3)	+1.45	[−0.58, +3.78]	10 / 5	0.3018	0.3018	`NULL_COMPATIBLE`
Secondary (A4 − A1)	+0.87	[−1.92, +3.66]	7 / 4	0.5488	0.5488	`NULL_COMPATIBLE`

Table 5: Downstream edit-location mini-test on SWE-bench Verified ($N = 100$, descriptive). ContextHit@5 marks gold presence in the 5 presented candidates; EditLoc@1/3 reports model prediction accuracy.

Arm	Candidate Context (Top-5)	ContextHit@5	EditLoc@1	EditLoc@3	Conversion
D0	BM25 Top-5	35.0%	33.0%	35.0%	0.943
D1	A4 Graph Top-5	37.0%	31.0%	37.0%	0.838
D2	A3 Control Top-5	38.0%	30.0%	37.0%	0.789
D3	Oracle Context (Transmission)	100.0%	**74.0%**	**85.0%**	0.740

+4.36 pp, both intervals lie entirely within the pre-registered equivalence bounds of $[-5.0, +5.0]$ pp, classifying both confirmatory contrasts as `NULL_COMPATIBLE` (verdict: `NOT_DEMONSTRATED`).

- **Lexical Seed Dominance & Rank Order**: Comparing graph expansion (A4) directly to seed reranking (A1, 46.2% Acc@5) yields an advantage of only 3 tasks out of 344 ($\Delta = +0.87$ pp, $n_{10} = 7, n_{01} = 4$, $p = 0.5488$), despite requiring 4,988 additional neural evaluations to score graph out-neighbors. Furthermore, un-reranked BM25 (B0) maintains higher MRR@50 (0.3439 vs 0.3409) and top-1 accuracy (25.3% vs 24.7%) than A4. Expanding candidate pools with non-graph controls (A2, A3) dilutes the scoring pool with distractors, lowering Acc@5 below A1; A4 appears superior to controls not due to topological discovery, but because graph pools suffer less distractor dilution.
- **Downstream Reader Transmission & Distractor Confusion** ($N = 100$): To determine whether local edge models exploit retrieved candidates, I evaluated arms D0–D3 under a 5-candidate prompt. In BM25 contexts (D0), ContextHit@5 is 35.0% and EditLoc@1 is 33.0% (conversion 94.3%), matching a trivial baseline that always selects candidate 1 (33.0%). In Graph contexts (D1), ContextHit@5 rises to 37.0%, but EditLoc@1 drops to 31.0% (conversion collapses to 83.8%), showing that structurally plausible graph neighbors (callers and callees) confuse the 2B reader model. Viability is demonstrated in the oracle arm (D3, randomized position), where Gemma 4 achieves 74.0% top-1 accuracy (67.5% chance-adjusted, 95% Wilson CI [64.6, 81.6]%, Table 5), proving that the retrieval channel (which supplies the gold target on only 35%–38% of tasks) is the primary agentic bottleneck.

5.6 Ablation Study

To evaluate the contribution of individual mechanisms under controlled conditions:

- **Without Specificity Ratio** ($\gamma = 0$): Perron at $\gamma = 0$ reduces identically to Standard PPR. Under diffuse linear teleportation on raw graphs ($N = 237$), unweighted diffusion collapses Function Acc@10 from 34.2% (81/237) to 10.1% (24/237, $\Delta = -24.1$ pp) as test-node contamination rises to 32.5%. On SWE-bench Verified ($N = 344$), removing specificity weighting drops HSI@10 from 99.5% to 90.8% (introducing 9.2% hub contamination), while top-10 accuracy remains at 50.3% vs. 50.0%.
- **Without Graph Sanitization (Raw vs. Sanitized)**: Retaining un-sanitized test subgraphs under diffuse priors funnels random-walk mass into testing harnesses, reducing Function Acc@10 by 3.8 pp (13.9% vs. 10.1%). Concentrating the teleport prior via top-25 softmax eliminates test-sink vulnera-

bility even without structural pruning.

6 Open-Source Ecosystem and Resource Deliverables

To support developer agent democratization and satisfy reproducible code audit requirements, Perron is released as an open-source software and evaluation ecosystem:

- **`perron-core` v0.3.0**: a Python package (`numpy`, `scipy`, `pyyaml`) providing zero-copy memory-mapped CSR ingestion, prior normalization, query-directed specificity diffusion, and AST editing, published on PyPI at https://pypi.org/project/perron-core/0.3.0/.
- **MCP server (`perron-mcp`)**: a JSON-RPC 2.0 stdio server (`perron.mcp_server`) for desktop agent environments with tools `retrieve_context`, `inspect_symbol_breadcrumbs`, and `build_code_graph`.
- **Dataset `swebench-lite-codegraphs`**: CSR edge topologies, dangling indicators, and stationary distributions for all 12 SWE-bench Lite repositories under Apache 2.0.
- **Dataset `swebench-strata-annotations`**: 800 per-instance lexical annotations across SWE-bench Lite and Verified (707 unique instances, 93 overlap) under CC0-1.0 at https://www.kaggle.com/datasets/yvliet/swebench-strata-annotations.
- **Dataset `gemma4-verified-reranking-scores`**: 19,298 scored candidate tuples (`scores.parquet`) and per-arm rankings for all 344 SWE-bench Verified tasks under Apache 2.0.
- **Kaggle Gemma 4 Workstation Kernel**: an interactive GPU demonstration notebook at https://www.kaggle.com/code/yvliet/perron-gemma-4-workstation.

7 Discussion and Limitations

While Perron acts as an invariant topological stabilizer under noisy priors and sustains robust hub suppression (99.5% HSI), several limitations remain: (1) static call graphs miss dynamic runtime dispatch, reflection (`getattr`), or monkey-patching; (2) across disconnected modules, diffusion relies entirely on the teleport prior; (3) structural diffusion alone does not outperform lexical BM25 on issue descriptions that explicitly quote target identifiers; (4) while reader transmission demonstrates 74.0% top-1 conversion on oracle contexts ($N = 100$), unconstrained generative patch synthesis was not evaluated, as small-model reflection loops require decoupled extraction budgets; and (5) the reference engine targets Python AST; while an initial polyglot Tree-sitter interface smoke test is implemented for TypeScript (`perron/polyglot/treesitter.py`) to verify CSR protocol extensibility, large-scale multilingual SWE-bench evaluation remains an open research direction.

8 Conclusion

On scale-free code graphs, Personalized PageRank collapse under default damping ($\beta = 0.85$) is driven by teleport prior dispersion and test-node sink absorption. Perron resolves this through post-walk Decoupled Specificity ($\pi_q/(\pi_g+\epsilon)^\gamma$) over static memory-mapped CSR operators. In a pre-specified confirmatory evaluation on SWE-bench Verified (344 callable tasks), Perron exhibits no statistically detectable difference from BM25 at top-10 (50.3% vs. 50.9%, non-inferiority inconclusive against a -2.0 pp margin). Pointwise reranking under Gemma 4 demonstrates that graph 1-hop expansion does not outperform lexical seed reranking (+0.87 pp, $p = 0.5488$), with un-reranked BM25 preserving superior rank order (MRR 0.3439 vs 0.3409), proving that retrieval candidate recall, rather than local reader discrimination, forms the decisive agentic bottleneck. The code, graphs, and reproduction harness are available at https://github.com/yvliet/perron and https://www.kaggle.com/code/yvliet/perron-gemma-4-workstation.

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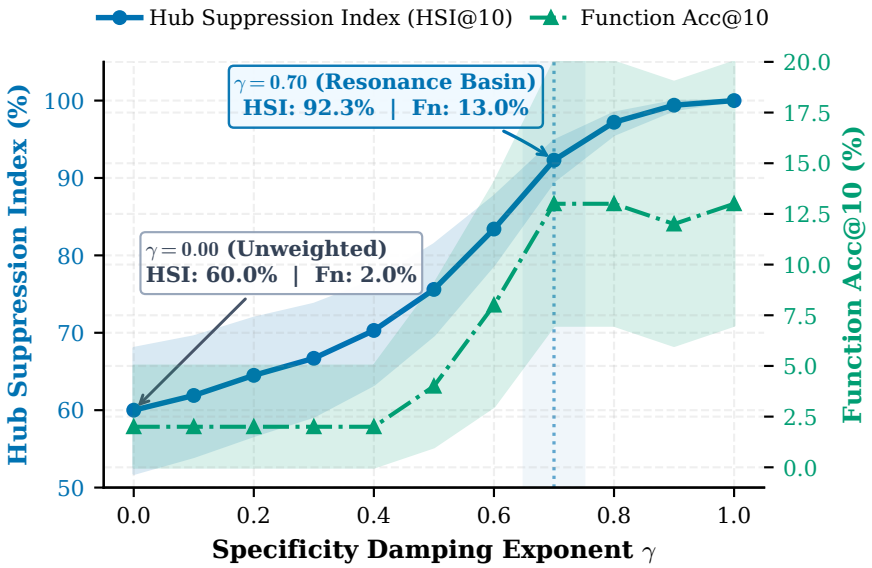


Figure 4: Specificity exponent sweep on dev_val ($N = 100$). Function Acc@10 and HSI@10 as a function of γ; $\gamma = 0$ is unweighted PPR. At the default $\gamma = 0.70$: Function Acc@10 13.0%, HSI@10 92.3%; at $\gamma = 0$: 2.0% and 60.0%.

A Supplementary Retrieval Tables

To evaluate retrieval behavior across varying degrees of textual specificity, Table 6 stratifies performance on SWE-bench Verified ($N = 344$ callable tasks) into three lexical accessibility strata: Stratum A where the gold target symbol is explicitly named in the problem description, Stratum B where the gold symbol is a direct neighbor of an issue mention, and Stratum C where the gold symbol is structurally distant without textual mention. Furthermore, Figure 4 plots the sensitivity of Function Acc@10 and HSI@10 across specificity exponent $\gamma \in [0, 1]$ on the development split ($N = 100$, sampled from SWE-bench Lite, disjoint from held-out evaluations).

Table 6: Retrieval performance stratified by lexical accessibility on SWE-bench Verified ($N = 344$ callable tasks). Stratum A: gold target symbol is explicitly referenced in the issue description ($N = 208$). Stratum B: gold symbol is 1-hop neighbor of an issue mention ($N = 5$). Stratum C: gold symbol is structurally distant without direct mention in the issue text ($N = 131$). Top-10 hits and accuracy rates are reported.

Method	**Stratum A** ($N = 208$)	**Stratum B** ($N = 5$)	**Stratum C** ($N = 131$)	**Overall** ($N = 344$)
Okapi BM25	147 (70.7%)	0 (0.0%)	28 (21.4%)	175 (50.9%)
Standard PPR ($\beta = 0.85$)	147 (70.7%)	0 (0.0%)	25 (19.1%)	172 (50.0%)
Degree-Normalized PPR	146 (70.2%)	0 (0.0%)	24 (18.3%)	170 (49.4%)
HippoRAG Prior Scaling	142 (68.3%)	0 (0.0%)	28 (21.4%)	170 (49.4%)
Perron Static ($\gamma = 0.70$)	146 (70.2%)	0 (0.0%)	27 (20.6%)	173 (50.3%)
Perron Adaptive	146 (70.2%)	0 (0.0%)	27 (20.6%)	173 (50.3%)
Slot Fusion (RRF)	147 (70.7%)	0 (0.0%)	27 (20.6%)	174 (50.6%)